

Remote Sensing Data Mining for Monitoring Frozen Peatland Ecosystems in the Hudson Bay Lowlands

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1 Context

Peatlands store more than a third of the global soil organic carbon stock. The Hudson Bay Lowlands, located in northern Canada, represent one of the largest continuous expanses of peatland between 60° and 70°N. In recent decades, this area has faced strong climate warming: air temperatures have increased at more than twice the global average, contributing to the rapid degradation of discontinuous permafrost.

In Wapusk National Park, peat plateau bogs form the dominant ecosystem, with variations in vegetation cover and surface moisture closely linked to underlying patterns of discontinuous permafrost, shaping a mosaic of distinct ecological zones. As thaw progresses, peat plateaus collapse, giving way to wetter environments colonized by Sphagnum mosses. These transitions result in major changes in carbon dynamics, methane fluxes, vegetation composition, and hydrology.

Remote sensing provides essential tools for monitoring these rapidly changing and spatially fragmented ecosystems. However, the fine-scale mosaic of dry, lichen-covered plateaus and waterlogged Sphagnum habitats presents challenges for conventional satellite-based classification. To address this, our study leverages ultra-high-resolution drone imagery alongside super-resolved Sentinel-2 time series data. This multi-scale approach enables detailed detection and mapping of subtle vegetation and hydrological changes across the landscape.

2 Bibliographic Synthesis

In the Hudson Bay Lowlands, multi-decadal satellite imagery (Sentinel, Landsat, ESA WorldCover) enables precise mapping of water surface changes, ecozone transitions, and lake desiccation processes linked to rising temperatures and permafrost thaw [1]. Spectral indices such as the Modified Normalized Difference Water Index (MNDWI) facilitate efficient discrimination of water bodies and quantification of hydrological changes.

At the vegetation level, Sphagnum mosses are central to peatland carbon and water cycling. Their large-scale monitoring is now possible through the integration of field and Sentinel-2 remote sensing data. Garisoain et al. [2] demonstrate that satellite-derived leaf area index (LAI) captures strong seasonal patterns in Sphagnum growth, correlating

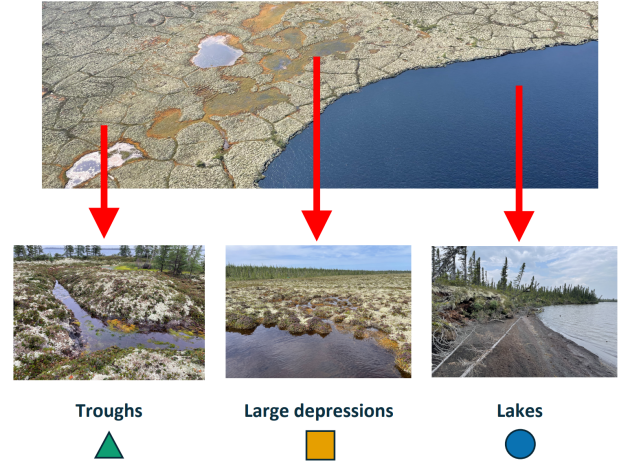


Figure 1: Peat plateau fragmented ecosystems in Wapusk National Park

with productivity, air temperature, and water table depth. The moisture soil index further refines the detection of Sphagnum-dominated areas and advances the representation of peatland vegetation in Earth system models.

Peatlands occupy just 3% of land area but store over a third of global soil carbon, and their stability is highly sensitive to moisture loss and disturbance. Degradation increases carbon emissions, accelerating climate warming. Remote sensing supports detailed assessments of peatland hydrology and vegetation, providing essential information for ecosystem conservation and restoration [3].

3 Objectives

This study aims to advance the monitoring and classification of frozen peatland ecosystems in the Hudson Bay Lowlands by integrating multi-scale remote sensing data, including high-resolution drone imagery and super-resolved satellite observations. The primary study area is the degraded peat plateau of Twin Lakes, near the Churchill research camp, where RGB, thermal, and digital surface model drone images are used and compared with Sentinel-2 data. The main objectives are to classify peatland ecozones at high resolution and to evaluate water surfaces and vegetation transitions in relation to permafrost thaw over the large areas covered by Sentinel-2 images. The methodology will then be extended to additional drone and satellite surveys in other study areas within Wapusk National Park. This

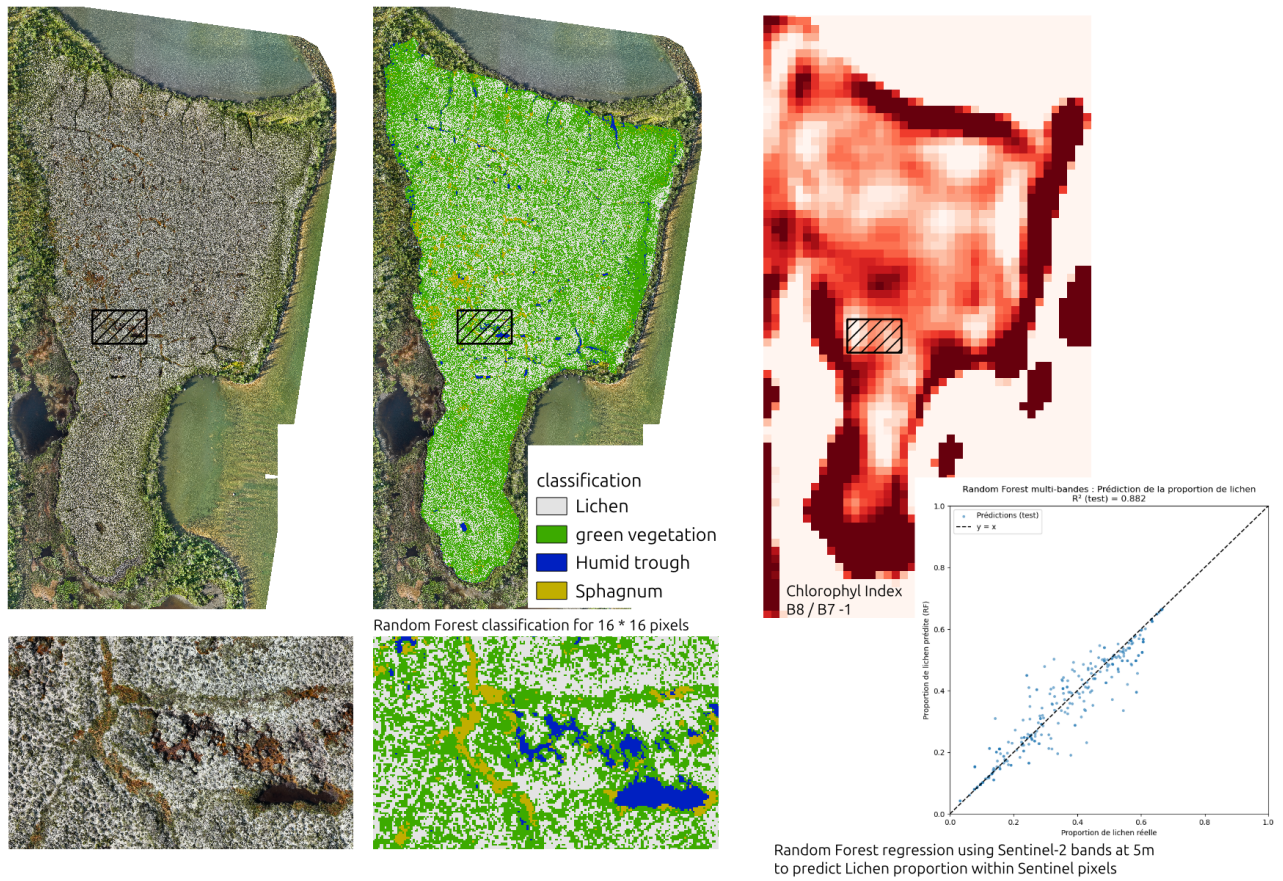


Figure 2: Random Forest classification of peat plateau surfaces and Sentinel-2 prediction of lichen cover. From left to right: drone RGB 1cm-resolution image, Random Forest Classification 16cm-resolution image, Sentinel-2 CI Index 5m-resolution. Scatterplot showing the relationship between predicted and observed lichen proportions ($R^2 = 0.882$).

work contributes to improving remote sensing approaches for Arctic ecosystem monitoring and provides valuable data to support ongoing studies on ecological transformations in peatlands.

4 Methods and Preliminary Results

To map the fine-scale heterogeneity of the Twin Lakes peat plateau, we used ultra-high-resolution drone imagery collected in July 2023. The dataset includes 1 cm-resolution visible images (RGB), digital surface model images (DSM, representing surface elevation), and 3 cm-resolution thermal images. These provide detailed spatial information essential for capturing the small-scale patterns of peat plateau surfaces (see Figure 1).

For classification, drone images were divided into 16×16 pixel tiles to preserve the microtopographic structure. We calculated statistical features for each tile and its neighbors to classify surface types. Hand-labeled samples were used to train a Random Forest (RF) classifier.

Initial attempts to include the adjacent forest were not effective due to complexity and limited relevance, so we focused the analysis on the peat plateau. Four main surface classes were retained for Twin Lakes: lichen, green vegetation, humid troughs, and Sphagnum. The results of this

classification are presented in Figure 2.

The classified drone map was upscaled to 5×5 m grids to match the Sentinel-2 satellite scale, allowing calculation of lichen cover per cell. For the regression, we used Sentinel-2 data acquired during summer 2023, taking the median value for each pixel. We included 10 spectral bands and 9 indices derived from these bands as predictors in the Random Forest regression, which predicted lichen proportion at this scale with a strong fit (see Figure 2).

Earlier tests with larger 32×32 tiles struggled to separate lichen from green vegetation, reducing regression accuracy. Adding a green vegetation class—much more common than Sphagnum at Twin Lakes—improved results by reducing noise in Sentinel-2 estimates. However, estimating trough proportions remains difficult at this site, so future work will target areas like Wapusk National Park, where Sphagnum and troughs are more abundant, to improve classification and regression.

In summary, these preliminary results demonstrate the value of combining drone and satellite data for detailed, scalable mapping of peatland surface types.

References

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